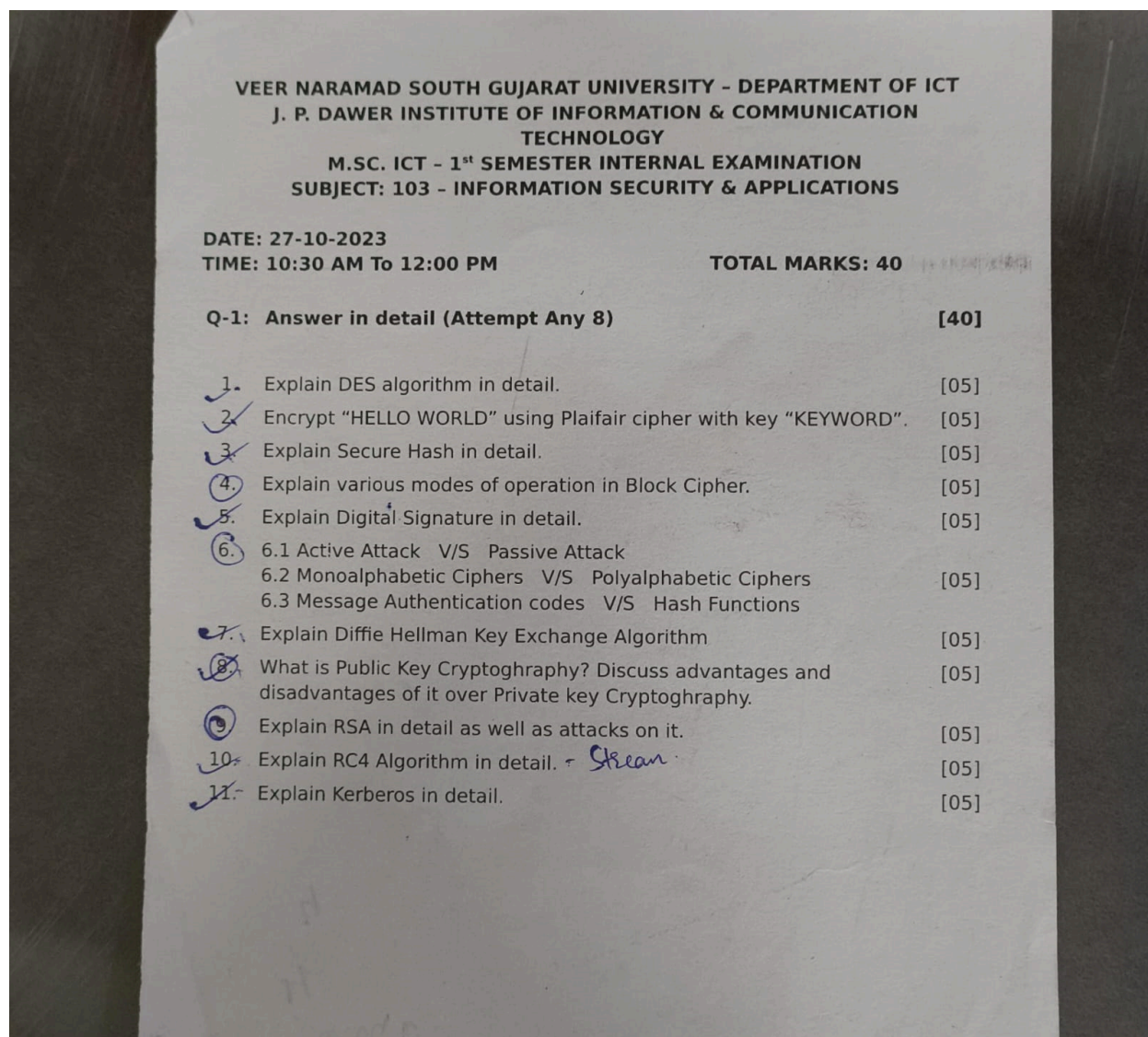




1. Explain DES algorithm in detail:

The **Data Encryption Standard (DES)** is a symmetric-key algorithm used for the encryption of digital data. Here's a breakdown of how DES works:

- **Key Size:** 64 bits (56 bits used for encryption, 8 bits used for parity).
- **Block Size:** 64 bits.
- **Rounds:** 16 rounds of permutation and substitution.
- **Process:**
 1. **Initial Permutation:** Rearranges the bits in the plaintext.

2. **Round Function:** Each round involves expansion, substitution (S-boxes), permutation (P-boxes), and XOR operation with a round key.
 3. **Subkeys:** 16 subkeys, each of 48 bits, are generated from the main key.
 4. **Final Permutation:** The output from the 16 rounds undergoes a final permutation to produce the ciphertext.
- **Strength:** DES has been considered weak due to its short key length, making it vulnerable to brute-force attacks. It is now replaced by more secure algorithms like AES.
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2. Encrypt "HELLO WORLD" using Playfair cipher with the key "KEYWORD":

Playfair Cipher is a digraph substitution cipher. Here's how to encrypt "HELLO WORLD" with the key "KEYWORD".

Key Matrix:

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	S
T	U	V	X	Z

- - **Steps:**
 1. Divide the plaintext "HELLO WORLD" into digraphs: "HE LL OW OR LD".
 2. Encrypt each pair:
 - **HE:** Both letters form a rectangle in the matrix; encrypted as **GA**.
 - **LL:** Same letter rule: insert an 'X'; becomes **LX**.
 - **OW:** Rectangle rule; encrypted as **CR**.
 - **OR:** Rectangle rule; encrypted as **KW**.
 - **LD:** Rectangle rule; encrypted as **AH**.
 3. Encrypted text: **GAXXCRKWXXAH** (Note: 'X' added for repeated letter or extra characters).
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3. Explain Secure Hash in detail:

Secure Hash Algorithms (SHA) are a family of cryptographic hash functions used to ensure data integrity. Some important points include:

- **Properties:**
 - **Deterministic:** Same input always produces the same hash output.

- **Fast:** Quick computation of hash values.
- **Pre-image Resistance:** Difficult to reverse-engineer the original message from the hash.
- **Collision Resistance:** Hard to find two different inputs with the same hash output.
- **Popular Algorithms:**
 - **SHA-1:** Produces a 160-bit hash value but is now considered insecure.
 - **SHA-256:** Produces a 256-bit hash value and is widely used in modern security.
 - **SHA-512:** Produces a 512-bit hash, used in systems that need higher security.

4. Explain various modes of operation in Block Cipher:

Block ciphers can operate in different modes, enhancing their versatility. Common modes include:

Mode	Description	Use Case
ECB	Electronic Codebook: Encrypts each block independently.	Small, independent data chunks
CBC	Cipher Block Chaining: Each block is XORed with the previous ciphertext block.	File encryption
CFB	Cipher Feedback: Converts block cipher into a stream cipher.	Real-time data encryption
OFB	Output Feedback: Converts block cipher into a stream cipher without chaining errors.	Wireless communication
CTR	Counter Mode: Each block is combined with a counter; allows parallel processing.	High-performance encryption

5. Explain Digital Signature in detail:

Digital Signature is a cryptographic mechanism that ensures:

1. **Authentication:** Verifies the sender's identity.

2. **Integrity:** Ensures the message hasn't been altered.
 3. **Non-repudiation:** Prevents the sender from denying the message.
 - **Working:**
 1. The sender creates a hash of the message.
 2. The hash is encrypted with the sender's **private key** to create the digital signature.
 3. The signature and the message are sent to the receiver.
 4. The receiver decrypts the signature using the sender's **public key**.
 5. The receiver hashes the message and compares it with the decrypted hash.
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6. Differences:

6.1 Active Attack vs Passive Attack:

Active Attack	Passive Attack
Alters the data being transmitted.	Monitors the data without altering it.
Examples: Modification, Denial of Service.	Examples: Eavesdropping, Traffic Analysis.
Easily detectable.	Difficult to detect.

6.2 Monoalphabetic Ciphers vs Polyalphabetic Ciphers:

Monoalphabetic Ciphers	Polyalphabetic Ciphers
Single substitution alphabet.	Multiple substitution alphabets.
Example: Caesar Cipher.	Example: Vigenère Cipher.
More vulnerable to frequency analysis attacks.	Less vulnerable to frequency analysis attacks.

6.3 Message Authentication Codes (MAC) vs Hash Functions:

Message Authentication Codes (MAC)	Hash Functions
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Provides message integrity and authentication.	Provides message integrity only.
Uses a secret key for verification.	No key is used; relies on hash value.

7. Explain Diffie Hellman Key Exchange Algorithm:

The **Diffie-Hellman Algorithm** allows two parties to generate a shared secret key over an insecure channel without transmitting the key.

- **Steps:**
 1. Both parties agree on public values: a prime number p and a base g .
 2. Each party generates a private key (a for Alice and b for Bob).
 3. Alice sends $g^a \text{ mod } p$ to Bob, and Bob sends $g^b \text{ mod } p$ to Alice.
 4. Both parties compute the shared secret key: Alice computes $(g^b)^a \text{ mod } p$, and Bob computes $(g^a)^b \text{ mod } p$.
 5. The result is the same for both parties, allowing them to use it as a shared key.

8. What is Public Key Cryptography? Discuss advantages and disadvantages:

Public Key Cryptography uses two keys: a **public key** for encryption and a **private key** for decryption.

- **Advantages:**
 1. Provides confidentiality, integrity, and authentication.
 2. Easier key management since public keys are openly distributed.
 3. No need to share the private key.
- **Disadvantages:**
 1. Slower than symmetric key cryptography.
 2. Requires more computational power and storage.

9. Explain RSA in detail as well as attacks on it:

RSA (Rivest–Shamir–Adleman) is a widely used public-key encryption algorithm.

- **Key Generation:**

1. Choose two large prime numbers, p and q .
 2. Compute $n = p * q$.
 3. Calculate Euler's totient function, $\phi(n) = (p-1)*(q-1)$.
 4. Choose a public exponent e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
 5. Compute the private key d , where $d * e \equiv 1 \pmod{\phi(n)}$.
- **Encryption:** Ciphertext $C = M^e \bmod n$.
 - **Decryption:** Plaintext $M = C^d \bmod n$.
 - **Attacks:**
 1. **Brute Force:** Trying all possible private keys.
 2. **Mathematical Attacks:** Factorizing n to find p and q .
 3. **Timing Attacks:** Measuring the time taken for decryption operations.
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10. Explain RC4 Algorithm in detail:

RC4 is a stream cipher used in secure protocols like SSL and WEP. It generates a pseudorandom stream of bits (keystream) and XORs it with plaintext.

- **Steps:**
 1. **Key-Scheduling Algorithm (KSA):** Initializes a 256-byte state array S using the key.
 2. **Pseudo-Random Generation Algorithm (PRGA):** Generates the keystream by permuting the state array.
 3. **Encryption:** XOR the keystream with the plaintext to produce ciphertext.
 - **Strength:** Fast and simple.
 - **Weakness:** Susceptible to biases in the keystream, leading to vulnerabilities in protocols like WEP.
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11. Explain Kerberos in detail:

Kerberos is a network authentication protocol that uses secret-key cryptography to provide secure authentication for client-server applications.

- **Components:**
 1. **Client:** The user or device requesting access.
 2. **Authentication Server (AS):** Verifies user credentials and issues a Ticket Granting Ticket (TGT).
 3. **Ticket Granting Server (TGS):** Issues service tickets based on the TGT.
 4. **Service Server (SS):** Provides the requested service using the service ticket.

- **Working:**
 1. The client sends a request to the AS.
 2. The AS authenticates the client and issues a TGT.
 3. The client uses the TGT to request a service ticket from the TGS.
 4. The client presents the service ticket to the SS for access.
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